PRODUCT USER MANUAL

For Global Ocean Multi Observation Products

MULTIOBS_GLO_PHY_MYNRT_015_003

Issue: 2.0

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2023/06	All	Creation of the document for the new product MULTIOBS_GLO_PHY_MYNRT_015_003 replacing MULTIOBS_GLO_PHY_NRT_015_003 and MULTIOBS_GLO_PHY_REP_015_004	H.Etienne	H.Etienne
2.0	2024/04		Update for November 2024 EIS : add tide current variables and related uncertainties.	H.Etienne	H.Etienne

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GLOSSARY AND ABBREVIATIONS

CF	Climate Forecast (convention for NetCDF)
Meridional Velocity	West to East component of the horizontal velocity vector
MOB	Multi OBservation
NetCDF	Network Common Data Form
NOAA	National Oceanic and Atmospheric Administration
NRT	Near Real Time
PU	Production Unit
Zonal Velocity	South to North component of the horizontal velocity vector

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

I.1 Summary

This guide describes the data product files from the MOB (Multi Observation) PU (Production Unit), what data services are available to access them, and how to use the files and services.

The product described here is **MULTIOBS_GLO_PHY_MYNRT_015_003** product. It contains **six datasets** on a 1/4° regular grid at a **1-hourly**, **daily** and at a **monthly** period:

- **cmems_obs-mob_glo_phy-cur_nrt_0.25deg_PT1H-i**: Near-Real Time 1-hourly (full temporal resolution) zonal and meridional components of the total velocity, tidal current and Ekman velocity fields.
- **cmems_obs-mob_glo_phy-cur_nrt_0.25deg_P1D-m**: Near-Real Time daily mean of the total, tide, Ekman and geostrophic currents, together with their respective daily uncertainties.
- **cmems_obs-mob_glo_phy-cur_nrt_0.25deg_P1M-m**: Near-Real Time monthly mean of the total, tide, Ekman and geostrophic currents, together with their respective monthly uncertainties.
- **cmems_obs-mob_glo_phy-cur_my_0.25deg_PT1H-i**: Reprocessed 1-hourly (full temporal resolution) zonal and meridional components of the total velocity, tidal current and Ekman velocity fields.
- **cmems_obs-mob_glo_phy-cur_my_0.25deg_P1D-m**: Reprocessed daily mean of the total, tide, Ekman and geostrophic currents, together with their respective daily uncertainties.
- **cmems_obs-mob_glo_phy-cur_my_0.25deg_P1M-m** : Reprocessed monthly mean of the total, tide, Ekman and geostrophic currents, together with their respective monthly uncertainties.

The three MY datasets are extended in time at least twice a year, so that they cover 1993 up to 6 to 18 months before present. The MY datasets benefit from the highest quality ocean observations inputs and are produced with a unique system (no change in time in the setup, extended in time using the same system) to minimize the risk of quality loss or spurious signals appearing in time.

The NRT hourly and daily datasets are updated daily with -1d, -3d and -6d fields to benefit from the updated NRT geostrophic component from the SL-TAC, while the monthly dataset is updated every 6th of the month.

Those products are a combination between satellite and in-situ data.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I.2 History of changes

November 2023	First release of **MULTIOBS_GLO_PHY_MYNRT_015_003 (new product) replacing MULTIOBS_GLO_PHY_NRT_015_003 and MULTIOBS_GLO_PHY_REP_015_004** Update of upstream data Increase MY time frequency from 3-hourly to 1-hourly Increase NRT time frequency from 6-hourly to 1-hourly Use of WIND_GLO_PHY_L4_NRT_012_004 wind fields
November 2024	Update of upstream data Add tidal current in the total velocity and in the files as separate component. Add tidal current uncertainties.

II DESCRIPTION OF THE PRODUCT SPECIFICATION

II.1 General Information

Table 1: MULTIOBS_GLO_PHY_MYNRT_015_0043 product specification customer name.

Product Lines	**MULTIOBS_GLO_PHY_MYNRT_015_003**
Geographical coverage	Global (89.875°S, 89.875°N; 179.875°W, 179.875°E) without the Mediterranean sea.
Variables	• Zonal and Meridional Total velocity • Zonal and Meridional Ekman velocity • Zonal and Meridional surface geostrophic velocity • Zonal and Meridional barotropic tidal current • Uncertainties on Zonal and Meridional Total velocity • Uncertainties on Zonal and Meridional Ekman velocity • Uncertainties on Zonal and Meridional surface geostrophic velocity • Uncertainties on Zonal and Meridional barotropic tidal current
Product Type	Multi Year Near Real Time
Available time series	MY datasets: 01/01/1993 – 6 to 18 months before present (see online catalogue). MY datasets are extended twice a year. NRT datasets: year-2 to day-1. NRT datasets are updated daily.
Temporal resolution	1-hourly fields (full resolution) Daily mean fields (centered on 12h00) Monthly mean fields (centered on the 15th of the month)
Target delivery time	16h UTC for NRT datasets.
Delivery mechanism	see section **Erreur ! Source du renvoi introuvable.**
Horizontal resolution	1/4° on a regular grid

Number of vertical levels	2 levels: surface and 15m depth.
Format	Netcdf CF4.0

NRT runtime schedule:

Every day, day-6, day-3 and day-1 hourly and daily nrt fields are produced.

They benefit from the last updated geostrophic current (SEALEVEL_GLO_PHY_L4_NRT_OBSERVATIONS_008_046) and day-1 wind fields (WIND_GLO_PHY_L4_NRT_012_004).

II.2 Details of datasets

Table 2 : list of the datasets and variable names and unit for MULTIOBS_GLO_PHY_MYNRT_015_003 product

MULTIOBS_GLO_PHY_MYNRT_015_003
Dataset: **cmems_obs-mob_glo_phy-cur_nrt_0.25deg_PT1H-i contains: nrt hourly instantaneous values** **cmems_obs-mob_glo_phy-cur_nrt_0.25deg_P1D-m contains: nrt daily mean values (24 hours average)** **cmems_obs-mob_glo_phy-cur_nrt_0.25deg_P1M-m contains: nrt monthly mean values** **cmems_obs-mob_glo_phy-cur_my_0.25deg_PT1H-i contains: my hourly instantaneous values** **cmems_obs-mob_glo_phy-cur_my_0.25deg_P1D-m contains: my daily mean values (24 hours average)** **cmems_obs-mob_glo_phy-cur_my_0.25deg_P1M-m contains: my monthly mean values** **Variables name in the NetCDF file and Unit: Long_name & Standard_name**
uo [m/s] **Absolute geostrophic velocity + depth Ekman + tide velocity: zonal component** **eastward_sea_water_velocity**
vo [m/s] **Absolute geostrophic velocity + depth Ekman + tide velocity: meridional component** **northward_sea_water_velocity**
ue[m/s] **Depth Ekman driven velocity : zonal component** **eastward_sea_water_velocity_due_to_ekman_drift**
ve[m/s] **Depth Ekman driven velocity : meridional component** **northward_sea_water_velocity_due_to_ekman_drift**
ugos[m/s] **Absolute surface geostrophic velocity : zonal component** **surface_geostrophic_eastward_sea_water_velocity**

MULTIOBS_GLO_PHY_MYNRT_015_003
vgos[m/s] **Absolute surface geostrophic velocity : meridional component** **surface_geostrophic_northward_sea_water_velocity**
utide[m/s] **Tide-induced velocity (Tide current): zonal component** **sea_water_x_velocity_due_to_tide**
vtide[m/s] **Tide-induced velocity (Tide current): meridional component** **sea_water_y_velocity_due_to_tide**
err_uo[m/s] **Error on absolute surface geostrophic + depth Ekman velocity + tide: zonal component** **eastward_sea_water_velocity_standard_error**
err_vo[m/s] **Error on absolute surface geostrophic + depth Ekman velocity + tide: meridional component** **northward_sea_water_velocity_standard_error**
err_ue[m/s] **Error on depth Ekman driven velocity : zonal component** **eastward_sea_water_velocity_due_to_ekman_drift_standard_error**
err_ve[m/s] **Error on depth Ekman driven velocity : meridional component** **northward_sea_water_velocity_due_to_ekman_drift_ standard_ error**
err_ugos[m/s] **Error on absolute surface geostrophic velocity: zonal component** **surface_geostophic_eastward_sea_water_velocity_standard_error**
err_vgos[m/s] **Error on absolute surface geostrophic velocity: meridional component** **surface_geostophic_northward_sea_water_velocity_standard_error**
err_utide[m/s] **Error on tide-induced velocity: zonal component** **sea_water_x_velocity_due_to_tide_standard_error**
err_vtide[m/s] **Error on tide-induced velocity: meridional component** **sea_water_y_velocity_due_to_tide_standard_error**

II.3 Production System Description

II.3.1 NRT production

The NRT datasets of **MULTIOBS_GLO_PHY_MYNRT_015_003** are provided by the CLS Production Center: **MULTIOBS-CLS-TOULOUSE-FR**.

Production is performed daily in 8 steps (Figure 1):

- Download the near real time (NRT) surface geostrophic current from the SEA LEVEL TAC altimetric product.
- Compute hourly tidal current on a ¼° spatial grid from FES2022 atlas (Carrere et al; 2022).
- Download and filter the Mixed Layer Depth (MLD) from MULTIOBS_GLO_PHY_NRT_015_012 product.
- Download the hourly stress equivalent 10m wind from WIND_GLO_PHY_L4_NRT_012_004 and interpolate on the ¼° output grid.
- Compute the surface and 15 m Ekman current using upgraded method from Mulet et al (2021), preprocessed MLD and wind stress equivalent.
- Add the geostrophic, tidal and Ekman components to compute gridded surface and 15 m total velocity fields.
- Compute daily total velocity error fields from MLD and preprocessed error parameters at the surface and 15m depth.
- Compute daily geostrophic, tide and Ekman error fields at the surface and 15m depth.

Ekman and geostrophic currents are computed on the same ¼° regular grid as the final total velocity.

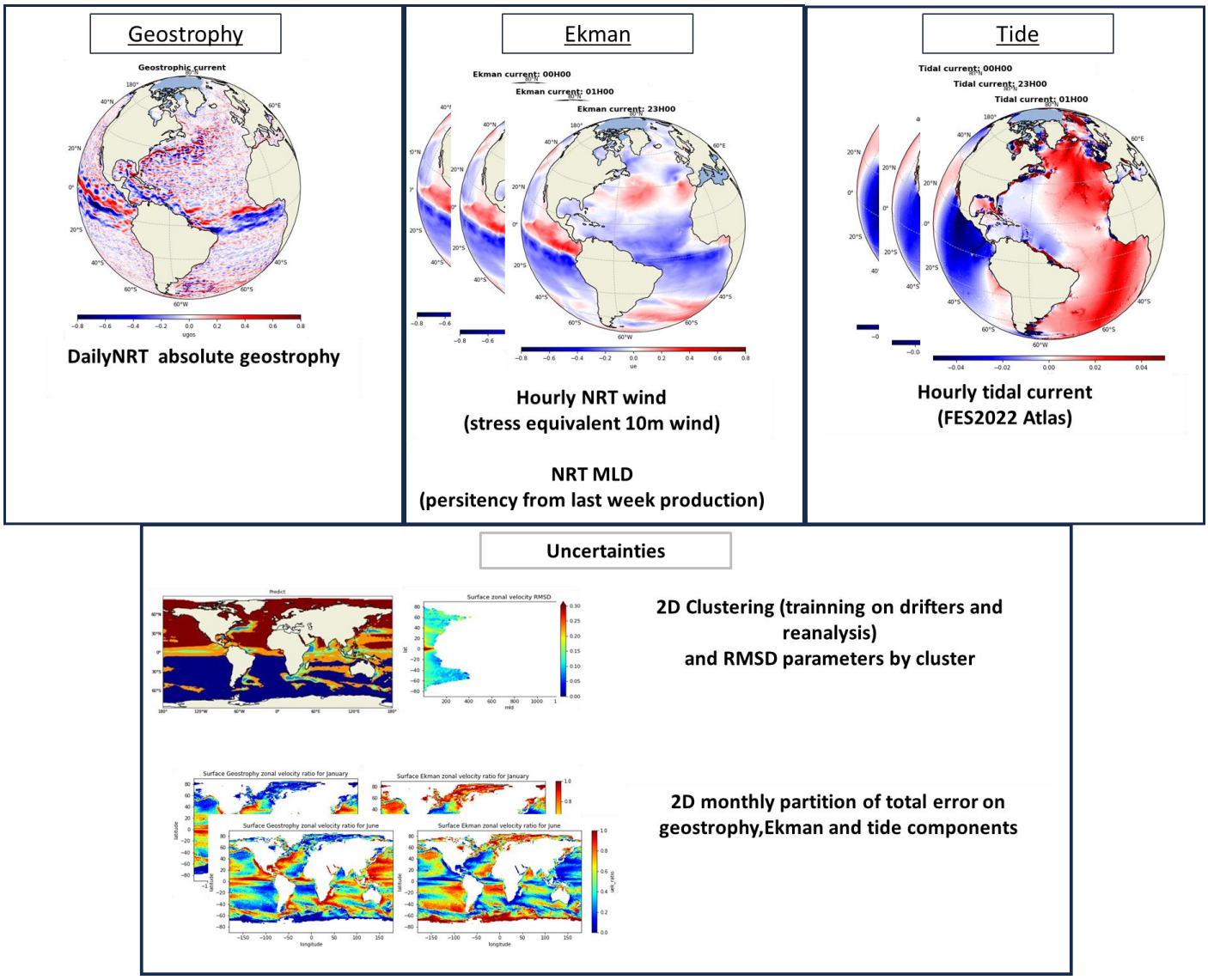

Figure 1 : Inputs of the MULTIOBS_GLO_PHY_MYNRT_015_003 system for a one-day NRT production.

II.3.2 MY production

The MY datasets production is performed daily in 8 steps (Figure 2):

- Get the MY surface geostrophic current from the SEA LEVEL TAC altimetric product.
- Compute hourly tidal current on a ¼° spatial grid from FES2022 atlas.
- Download and filter the Mixed Layer Depth (MLD) from dataset-armor-3d-rep-weekly of MULTIOBS_GLO_PHY_MYNRT_015_012 product. MLD is then interpolated daily in time.
- Download the hourly ERA5 wind stress and interpolate on the ¼° output grid.

- Compute the surface and 15 m Ekman current using upgraded method from Mulet et al (2021), preprocessed MLD and wind stress.
- Add the geostrophic, tidal and Ekman components to compute gridded surface and 15 m total velocity fields.
- Compute daily total velocity error fields from MLD and preprocessed error parameters at the surface and 15m depth.
- Compute daily geostrophic, tide and Ekman error fields at the surface and 15m depth.

Ekman and geostrophic currents are computed on the same ¼° regular grid as the final total velocity.

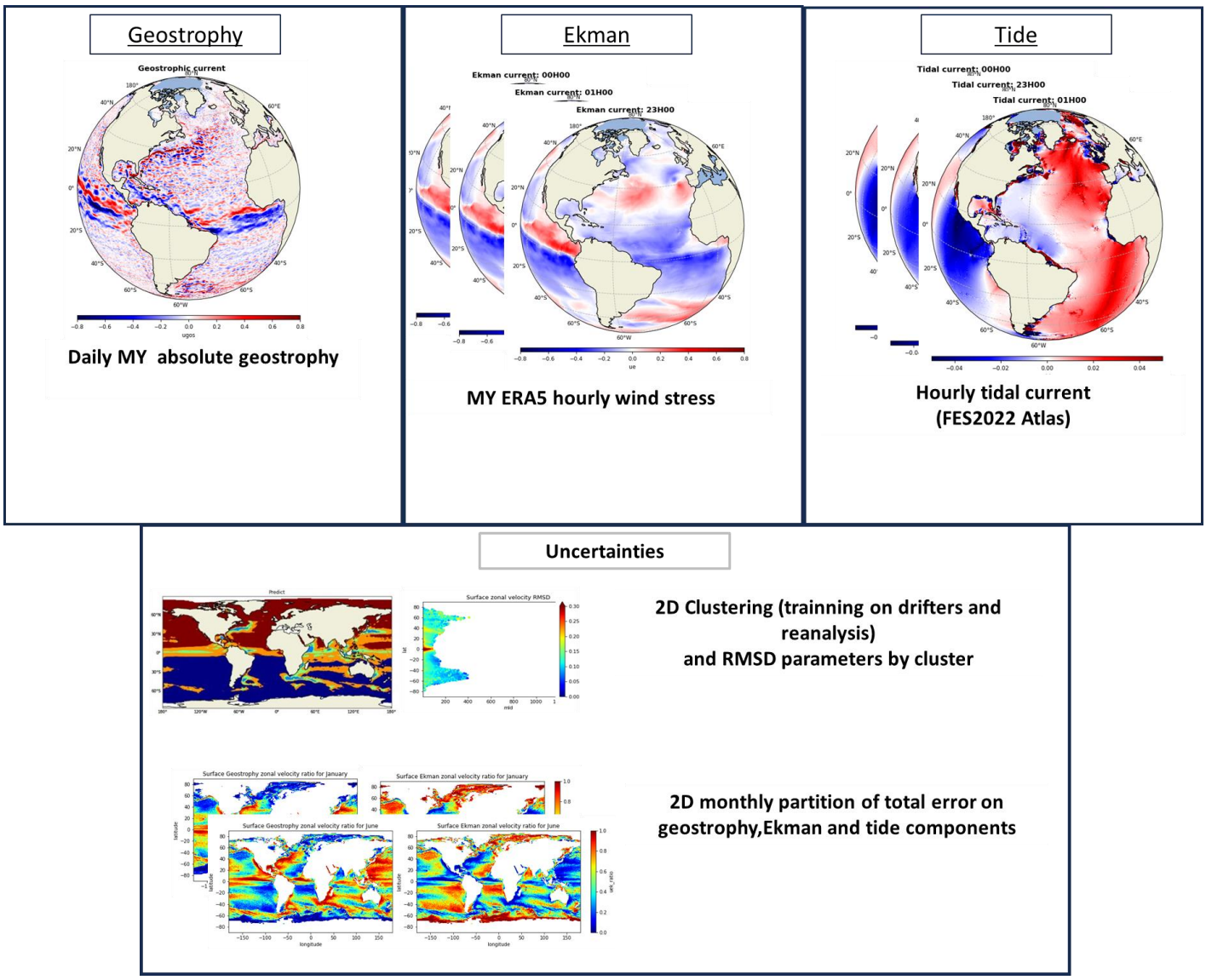

Figure 2 : Inputs of the MULTIOBS_GLO_PHY_MYNRT_015_003 system for a one-day MY production.

II.4 Grid

All products are delivered on the same 0.25° x 0.25° spatial grid.

II.5 Vertical Levels

Except geostrophy (only surface field) and tide (barotropic field), all the variables are provided at the surface and 15m depth.

II.6 Processing information

II.6.1 Update Time

NRT datasets are updated daily with a target delivery time at 16h00 UTC.

II.6.2 Temporal extent of MY product

They are extended twice a year.

II.6.3 Time averaging

cmems_obs-mob_glo_phy-cur_nrt_0.25deg_P1D-m and **cmems_obs-mob_glo_phy-cur_my_0.25deg_P1D-m** are daily mean datasets (averaged over 24h) and centered on 12h00.

cmems_obs-mob_glo_phy-cur_nrt_0.25deg_P1M-m and **cmems_obs-mob_glo_phy-cur_my_0.25deg_P1M-m** are monthly mean values centered on the 15th of the month.

III FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:

What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:

how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

IV.1 Nomenclature of files when downloaded through the Copernicus Marine Web Portal Subset Service

Files nomenclature when downloaded through the Web Portal Subset is based on product dataset name and a numerical reference related to the request date on the MIS.

The scheme is: **datasetname_nnnnnn.nc**

where :

.datasetname is a character string within one of the following :

- cmems_obs-mob_glo_phy-cur_my_0.25deg_PT1H-i
- cmems_obs-mob_glo_phy-cur_nrt_0.25deg_PT1H-i
- cmems_obs-mob_glo_phy-cur_my_0.25deg_P1D-m
- cmems_obs-mob_glo_phy-cur_nrt_0.25deg_P1D-m
- cmems_obs-mob_glo_phy-cur_my_0.25deg_P1M-m
- cmems_obs-mob_glo_phy-cur_nrt_0.25deg_P1M-m

. nnnnnn: 13 digit integer corresponding to the current time (download time) in milliseconds since January 1, 1970 midnight UTC.

.nc: standard NetCDF filename extension.

Example: dataset-uv-rep-hourly_1318943720287.nc

IV.2 Nomenclature of original files

MULTIOBS_GLO_PHY_MYNRT_015_003 files nomenclature when downloaded through the FTP is based as follows:

IV.2.1 Dataset cmems_obs-mob_glo_phy-cur_my_0.25deg_PT1H-i

dataset-uv-rep-hourly_YYYYMMDDT0000Z_PYYYYMMDDThhmmZ.nc

IV.2.2 Dataset cmems_obs-mob_glo_phy-cur_my_0.25deg_P1D-m

dataset-uv-rep-daily_YYYYMMDDT1200Z_PYYYYMMDDThhmmZ.nc

IV.2.3 Dataset cmems_obs-mob_glo_phy-cur_my_0.25deg_P1M-m

dataset-uv-rep-monthly_YYYYMMT0000Z_PYYYYMMDDThhmmZ.nc

IV.2.4 Dataset cmems_obs-mob_glo_phy-cur_nrt_0.25deg_PT1H-i

dataset-uv-nrt-hourly_YYYYMMDDT0000Z_PYYYYMMDDThhmmZ.nc

IV.2.5 Dataset cmems_obs-mob_glo_phy-cur_nrt_0.25deg_P1D-m

dataset-uv-nrt-daily_YYYYMMDDT1200Z_PYYYYMMDDThhmmZ.nc

IV.2.6 Dataset cmems_obs-mob_glo_phy-cur_nrt_0.25deg_P1M-m

dataset-uv-nrt-monthly_YYYYMMT0000Z_PYYYYMMDDThhmmZ.nc

IV.3 Other information: mean centre of Products, land mask value, missing value

Real_Value = (Display_Value X scale_factor) + add_offset

IV.3.1 File size

DATASET NAME	FILE NAME	DIMENSION [MB]	
		Compressed	Uncompressed
cmems_obs-mob_glo_phy-cur_my_0.25deg_PT1H-i	dataset-uv-rep-hourly_YYYYMMDDT0000Z_PYYYYMMDDThhmmZ.nc		475
cmems_obs-mob_glo_phy-cur_my_0.25deg_P1D-m	dataset-uv-rep-daily_YYYYMMDDT1200Z_PYYYYMMDDThhmmZ.nc		56
cmems_obs-mob_glo_phy-cur_my_0.25deg_P1M-m	dataset-uv-rep-monthly_YYYYMMT0000Z_PYYYYMMDDThhmmZ.nc		56
cmems_obs-mob_glo_phy-cur_nrt_0.25deg_PT1H-i	dataset-uv-nrt-hourly_YYYYMMDDT0000Z_PYYYYMMDDThhmmZ.nc		475
cmems_obs-mob_glo_phy-cur_nrt_0.25deg_P1D-m	dataset-uv-nrt-daily_YYYYMMDDT1200Z_PYYYYMMDDThhmmZ.nc		56
cmems_obs-mob_glo_phy-cur_nrt_0.25deg_P1M-m	dataset-uv-nrt-monthly_YYYYMMT0000Z_PYYYYMMDDThhmmZ.nc		56

IV.4 Structure and semantic of netCDF maps files

Examples of the header of output NetCDF files are inserted in annex, for each dataset.

V REFERENCES

Mulet, S., Rio, M.-H., Etienne, H., Artana, C., Cancet, M., Dibarboure, G., Feng, H., Husson, R., Picot, N., Provost, C., and Strub, P. T., 2021: The new CNES-CLS18 global mean dynamic topography, Ocean Sci., 17, 789-808

Carrere L., Lyard F., Cancet M., Allain D., Fouchet E., Dabat. M, Tchilibou M., Ferrari R. and Faugere Y.: new FES2022 Tidal atlas, OSTST 2022

VI ANNEX

VI.1.1 dataset-uv-rep-hourly_*YYYYMMDDT0000Z_PYYYYMMDDThhmmZ*.nc and dataset-uv-nrt-hourly_*YYYYMMDDT0000Z_PYYYYMMDDThhmmZ*.nc

```
dimensions:
    time = 24 ;
    latitude = 720 ;
    longitude = 1440 ;
    depth = 2 ;
variables:
    int crs ;
        crs:grid_mapping_name = "latitude_longitude" ;
        crs:semi_major_axis = 6378137LL ;
        crs:inverse_flattening = 298.257223563 ;
    float time(time) ;
        time:units = "hours since 1950-01-01 00:00:00 UTC" ;
        time:long_name = "Time" ;
        time:standard_name = "time" ;
        time:calendar = "gregorian" ;
        time:axis = "T" ;
    float latitude(latitude) ;
        latitude:long_name = "Latitude" ;
        latitude:standard_name = "latitude" ;
        latitude:units = "degrees_north" ;
        latitude:axis = "Y" ;
    float longitude(longitude) ;
        longitude:standard_name = "longitude" ;
        longitude:long_name = "Longitude" ;
        longitude:units = "degrees_east" ;
        longitude:axis = "X" ;
    short depth(depth) ;
        depth:standard_name = "depth" ;
```

```
        depth:long_name = "Depth" ;
        depth:units = "m" ;
        depth:axis = "Z" ;
        depth:positive = "down" ;
    short uo(time, depth, latitude, longitude) ;
        uo:_FillValue = 32767s ;
        uo:units = "m/s" ;
        uo:long_name = "Absolute geostrophic velocity + depth Ekman + tide velocity: zonal component" ;
        uo:standard_name = "eastward_sea_water_velocity" ;
        uo:grid_mapping = "crs" ;
        uo:scale_factor = 0.001 ;
    short vo(time, depth, latitude, longitude) ;
        vo:_FillValue = 32767s ;
        vo:units = "m/s" ;
        vo:long_name = "Absolute geostrophic velocity + depth Ekman + tide velocity: meridional component" ;
        vo:standard_name = "northward_sea_water_velocity" ;
        vo:grid_mapping = "crs" ;
        vo:scale_factor = 0.001 ;
    short ue(time, depth, latitude, longitude) ;
        ue:_FillValue = 32767s ;
        ue:units = "m/s" ;
        ue:long_name = "Depth Ekman driven velocity : zonal component" ;
        ue:standard_name = "eastward_sea_water_velocity_due_to_ekman_drift" ;
        ue:grid_mapping = "crs" ;
        ue:scale_factor = 0.001 ;
    short ve(time, depth, latitude, longitude) ;
        ve:_FillValue = 32767s ;
        ve:units = "m/s" ;
        ve:long_name = "Depth Ekman driven velocity : meridional component" ;
        ve:standard_name = "northward_sea_water_velocity_due_to_ekman_drift" ;
        ve:grid_mapping = "crs" ;
        ve:scale_factor = 0.001 ;
    short utide(time, latitude, longitude) ;
        utide:_FillValue = 32767s ;
        utide:units = "m/s" ;
```

```
        utide:long_name = "Tide-induced velocity (Tide current): zonal component" ;
        utide:standard_name = "sea_water_x_velocity_due_to_tide" ;
        utide:grid_mapping = "crs" ;
        utide:scale_factor = 0.001 ;
    short vtide(time, latitude, longitude) ;
        vtide:_FillValue = 32767s ;
        vtide:units = "m/s" ;
        vtide:long_name = "Tide-induced velocity (Tide current): meridional component" ;
        vtide:standard_name = "sea_water_y_velocity_due_to_tide" ;
        vtide:grid_mapping = "crs" ;
        vtide:scale_factor = 0.001 ;

// global attributes:
                :geospatial_lon_units = "degrees_east" ;
                :geospatial_vertical_max = 0. ;
                :product_version = "4.0" ;
                :geospatial_lat_units = "degrees_north" ;
                :creation_date = "2024-04-22 10:59:25" ;
                :contact = "servicedesk.cmems@mercator-ocean.eu" ;
                :references = "Mulet S., Rio M.-H., Etienne H., Artana C., Cancet M., Dibarboure G., Feng H., Husson R., Picot N., Provost C. and Strub P. T.: The new CNES-CLS18 global mean dynamic topography, Ocean Sci., 17, 789-808;  Carrere L., Lyard F., Cancet M., Allain D., Fouchet E., Dabat. M, Tchilibou M., Ferrari R. and Faugere Y.: new FES2022 Tidal atlas, OSTST 2022." ;
                :geospatial_vertical_min = -15. ;
                :Conventions = "CF-1.6" ;
                :institution = "CLS" ;
                :lon_min = -179.875 ;
                :lon_max = 179.875 ;
                :lon_res = 0.25 ;
                :lat_min = -89.875 ;
                :lat_max = 89.875 ;
                :lat_res = 0.25 ;
                :title = "Total surface and 15m velocities" ;
```

VI.1.2 dataset-uv-rep-daily_YYYYMMDDT1200Z_PYYYYMMDDThhmmZ.nc and dataset-uv-nrt-daily_YYYYMMDDT1200Z_PYYYYMMDDThhmmZ.nc

```
dimensions:
    latitude = 720 ;
    longitude = 1440 ;
    depth = 2 ;
    time = 1 ;
variables:
    int crs ;
        crs:grid_mapping_name = "latitude_longitude" ;
        crs:semi_major_axis = 6378137LL ;
        crs:inverse_flattening = 298.257223563 ;
    float latitude(latitude) ;
        latitude:long_name = "Latitude" ;
        latitude:standard_name = "latitude" ;
        latitude:units = "degrees_north" ;
        latitude:axis = "Y" ;
    float longitude(longitude) ;
        longitude:standard_name = "longitude" ;
        longitude:long_name = "Longitude" ;
        longitude:units = "degrees_east" ;
        longitude:axis = "X" ;
    short depth(depth) ;
        depth:standard_name = "depth" ;
        depth:long_name = "Depth" ;
        depth:units = "m" ;
        depth:axis = "Z" ;
        depth:positive = "down" ;
    float time(time) ;
        time:units = "hours since 1950-01-01 00:00:00 UTC" ;
        time:long_name = "Time" ;
        time:standard_name = "time" ;
        time:calendar = "gregorian" ;
        time:axis = "T" ;
    short uo(time, depth, latitude, longitude) ;
        uo:_FillValue = 32767s ;
        uo:units = "m/s" ;
```

```
        uo:long_name = "Absolute geostrophic velocity + depth Ekman + tide velocity: zonal component" ;
        uo:standard_name = "eastward_sea_water_velocity" ;
        uo:grid_mapping = "crs" ;
        uo:scale_factor = 0.001 ;
    short vo(time, depth, latitude, longitude) ;
        vo:_FillValue = 32767s ;
        vo:units = "m/s" ;
        vo:long_name = "Absolute geostrophic velocity + depth Ekman + tide velocity: meridional component" ;
        vo:standard_name = "northward_sea_water_velocity" ;
        vo:grid_mapping = "crs" ;
        vo:scale_factor = 0.001 ;
    short ue(time, depth, latitude, longitude) ;
        ue:_FillValue = 32767s ;
        ue:units = "m/s" ;
        ue:long_name = "Depth Ekman driven velocity : zonal component" ;
        ue:standard_name = "eastward_sea_water_velocity_due_to_ekman_drift" ;
        ue:grid_mapping = "crs" ;
        ue:scale_factor = 0.001 ;
    short ve(time, depth, latitude, longitude) ;
        ve:_FillValue = 32767s ;
        ve:units = "m/s" ;
        ve:long_name = "Depth Ekman driven velocity : meridional component" ;
        ve:standard_name = "northward_sea_water_velocity_due_to_ekman_drift" ;
        ve:grid_mapping = "crs" ;
        ve:scale_factor = 0.001 ;
    short utide(time, latitude, longitude) ;
        utide:_FillValue = 32767s ;
        utide:units = "m/s" ;
        utide:long_name = "Tide-induced velocity (Tide current): zonal component" ;
        utide:standard_name = "sea_water_x_velocity_due_to_tide" ;
        utide:grid_mapping = "crs" ;
        utide:scale_factor = 0.001 ;
    short vtide(time, latitude, longitude) ;
        vtide:_FillValue = 32767s ;
        vtide:units = "m/s" ;
```

vtide:long_name = "Tide-induced velocity (Tide current): meridional component" ;

vtide:standard_name = "sea_water_y_velocity_due_to_tide" ;

vtide:grid_mapping = "crs" ;

vtide:scale_factor = 0.001 ;

short ugos(time, latitude, longitude) ;

ugos:_FillValue = 32767s ;

ugos:units = "m/s" ;

ugos:long_name = "Absolute surface geostrophic velocity : zonal component" ;

ugos:standard_name = "surface_geostrophic_eastward_sea_water_velocity" ;

ugos:scale_factor = 0.001 ;

short vgos(time, latitude, longitude) ;

vgos:_FillValue = 32767s ;

vgos:units = "m/s" ;

vgos:long_name = "Absolute surface geostrophic velocity : meridional component" ;

vgos:standard_name = "surface_geostrophic_northward_sea_water_velocity" ;

vgos:scale_factor = 0.001 ;

short err_uo(time, depth, latitude, longitude) ;

err_uo:_FillValue = 32767s ;

err_uo:units = "m/s" ;

err_uo:standard_name = "eastward_sea_water_velocity_standard_error" ;

err_uo:long_name = " Error on absolute surface geostrophic + depth Ekman velocity + tide: zonal component";

err_uo:grid_mapping = "crs" ;

err_uo:scale_factor = 0.001 ;

short err_vo(time, depth, latitude, longitude) ;

err_vo:_FillValue = 32767s ;

err_vo:units = "m/s" ;

err_vo:standard_name = "northward_sea_water_velocity_standard_error" ;

err_vo:long_name = " Error on absolute surface geostrophic + depth Ekman velocity + tide: meridional component" ;

err_vo:grid_mapping = "crs" ;

err_vo:scale_factor = 0.001 ;

short err_ue(time, depth, latitude, longitude) ;

err_ue:_FillValue = 32767s ;

err_ue:units = "m/s" ;

err_ue:standard_name = "eastward_sea_water_velocity_due_to_ekman_drift_standard_error" ;

```
        string err_ue:long_name = "Error on depth Ekman driven velocity : zonal component" ;
        err_ue:grid_mapping = "crs" ;
        err_ue:scale_factor = 0.001 ;
    short err_ve(time, depth, latitude, longitude) ;
        err_ve:_FillValue = 32767s ;
        err_ve:units = "m/s" ;
        err_ve:standard_name = "northward_sea_water_velocity_due_to_ekman_drift_ standard_ error" ;
        string err_ve:long_name = "Error on depth Ekman driven velocity : meridional component" ;
        err_ve:grid_mapping = "crs" ;
        err_ve:scale_factor = 0.001 ;
    short err_ugos(time, depth, latitude, longitude) ;
        err_ugos:_FillValue = 32767s ;
        err_ugos:units = "m/s" ;
        err_ugos:standard_name = "surface_geostophic_eastward_sea_water_velocity_standard_error" ;
        err_ugos:long_name = "Error on absolute surface geostrophic velocity: zonal component " ;
        err_ugos:grid_mapping = "crs" ;
        err_ugos:scale_factor = 0.001 ;
    short err_vgos(time, depth, latitude, longitude) ;
        err_vgos:_FillValue = 32767s ;
        err_vgos:units = "m/s" ;
        err_vgos:standard_name = "surface_geostophic_northward_sea_water_velocity_standard_error" ;
        err_vgos:long_name = "Error on absolute surface geostrophic velocity: meridional component " ;
        err_vgos:grid_mapping = "crs" ;
        err_vgos:scale_factor = 0.001 ;
    short err_utide(time, depth, latitude, longitude) ;
        err_utide:_FillValue = 32767s ;
        err_utide:units = "m/s" ;
        err_utide:standard_name = "sea_water_x_velocity_due_to_tide_standard_error" ;
        err_utide:long_name = "Error on tide-induced velocity: zonal component " ;
        err_utide:grid_mapping = "crs" ;
        err_utide:scale_factor = 0.001 ;
    short err_vtide(time, depth, latitude, longitude) ;
        err_vtide:_FillValue = 32767s ;
        err_vtide:units = "m/s" ;
        err_vtide:standard_name = "sea_water_y_velocity_due_to_tide_velocity_standard_error" ;
```

err_vtide:long_name = "Error on tide-induced velocity: meridional component " ;

err_vtide:grid_mapping = "crs" ;

err_vtide:scale_factor = 0.001 ;

// global attributes:

:geospatial_lon_units = "degrees_east" ;

:geospatial_vertical_max = 0. ;

:product_version = "4.0" ;

:geospatial_lat_units = "degrees_north" ;

:creation_date = "2024-04-22 13:09:14" ;

:contact = "servicedesk.cmems@mercator-ocean.eu" ;

:references = "Mulet S., Rio M.-H., Etienne H., Artana C., Cancet M., Dibarboure G., Feng H., Husson R., Picot N., Provost C. and Strub P. T.: The new CNES-CLS18 global mean dynamic topography, Ocean Sci., 17, 789-808; Carrere L., Lyard F., Cancet M., Allain D., Fouchet E., Dabat. M, Tchilibou M., Ferrari R. and Faugere Y.: new FES2022 Tidal atlas, OSTST 2022." ;

:geospatial_vertical_min = -15. ;

:Conventions = "CF-1.6" ;

:institution = "CLS" ;

:lon_min = -179.875f ;

:lon_max = 179.875f ;

:lon_res = 0.25 ;

:lat_min = -89.875f ;

:lat_max = 89.875f ;

:lat_res = 0.25 ;

:title = "Daily mean total surface and 15m velocities" ;

VI.1.3 dataset-uv-rep-monthly_YYYYMMT0000Z_PYYYYMMDDThhmmZ.nc and dataset-uv-nrt-monthly_YYYYMMT0000Z_PYYYYMMDDThhmmZ.nc

dimensions:

longitude = 1440 ;

latitude = 720 ;

depth = 2 ;

time = 1 ;

```
variables:
    int crs ;
        crs:grid_mapping_name = "latitude_longitude" ;
        crs:semi_major_axis = 6378137LL ;
        crs:inverse_flattening = 298.257223563 ;
    float longitude(longitude) ;
        longitude:standard_name = "longitude" ;
        longitude:long_name = "Longitude" ;
        longitude:units = "degrees_east" ;
        longitude:axis = "X" ;
    float latitude(latitude) ;
        latitude:long_name = "Latitude" ;
        latitude:standard_name = "latitude" ;
        latitude:units = "degrees_north" ;
        latitude:axis = "Y" ;
    short depth(depth) ;
        depth:standard_name = "depth" ;
        depth:long_name = "Depth" ;
        depth:units = "m" ;
        depth:axis = "Z" ;
        depth:positive = "down" ;
    float time(time) ;
        time:units = "hours since 1950-01-01 00:00:00 UTC" ;
        time:long_name = "Time" ;
        time:standard_name = "time" ;
        time:calendar = "gregorian" ;
        time:axis = "T" ;
    short uo(time, depth, latitude, longitude) ;
        uo:_FillValue = 32767s ;
        uo:units = "m/s" ;
        uo:long_name = " Absolute geostrophic velocity + depth Ekman + tide velocity: zonal component" ;
        uo:standard_name = "eastward_sea_water_velocity" ;
        uo:grid_mapping = "crs" ;
        uo:scale_factor = 0.001 ;
    short vo(time, depth, latitude, longitude) ;
```

```
        vo:_FillValue = 32767s ;
        vo:units = "m/s" ;
        vo:long_name = " Absolute geostrophic velocity + depth Ekman + tide velocity: meridional component" ;
        vo:standard_name = "northward_sea_water_velocity" ;
        vo:grid_mapping = "crs" ;
        vo:scale_factor = 0.001 ;
    short ue(time, depth, latitude, longitude) ;
        ue:_FillValue = 32767s ;
        ue:units = "m/s" ;
        ue:long_name = "Depth Ekman driven velocity : zonal component" ;
        ue:standard_name = "eastward_sea_water_velocity_due_to_ekman_drift" ;
        ue:grid_mapping = "crs" ;
        ue:scale_factor = 0.001 ;
    short ve(time, depth, latitude, longitude) ;
        ve:_FillValue = 32767s ;
        ve:units = "m/s" ;
        ve:long_name = "Depth Ekman driven velocity : meridional component" ;
        ve:standard_name = "northward_sea_water_velocity_due_to_ekman_drift" ;
        ve:grid_mapping = "crs" ;
        ve:scale_factor = 0.001 ;
    short utide(time, latitude, longitude) ;
        utide:_FillValue = 32767s ;
        utide:units = "m/s" ;
        utide:long_name = "Tide-induced velocity (Tide current): zonal component" ;
        utide:standard_name = "sea_water_x_velocity_due_to_tide" ;
        utide:grid_mapping = "crs" ;
        utide:scale_factor = 0.001 ;
    short vtide(time, latitude, longitude) ;
        vtide:_FillValue = 32767s ;
        vtide:units = "m/s" ;
        vtide:long_name = "Tide-induced velocity (Tide current): meridional component" ;
        vtide:standard_name = "sea_water_y_velocity_due_to_tide" ;
        vtide:grid_mapping = "crs" ;
        vtide:scale_factor = 0.001 ;
    short ugos(time, latitude, longitude) ;
```

```
        ugos:_FillValue = 32767s ;
        ugos:units = "m/s" ;
        ugos:long_name = "Absolute surface geostrophic velocity : zonal component" ;
        ugos:standard_name = "surface_geostrophic_eastward_sea_water_velocity" ;
        ugos:scale_factor = 0.001 ;
    short vgos(time, latitude, longitude) ;
        vgos:_FillValue = 32767s ;
        vgos:units = "m/s" ;
        vgos:long_name = "Absolute surface geostrophic velocity : meridional component" ;
        vgos:standard_name = "surface_geostrophic_northward_sea_water_velocity" ;
        vgos:scale_factor = 0.001
    short err_uo(time, depth, latitude, longitude) ;
        err_uo:_FillValue = 32767s ;
        err_uo:units = "m/s" ;
        err_uo:standard_name = "eastward_sea_water_velocity_standard_error" ;
        err_uo:long_name = " Error on absolute surface geostrophic + depth Ekman velocity + tide: zonal component" ;
        err_uo:grid_mapping = "crs" ;
        err_uo:scale_factor = 0.001 ;
    short err_vo(time, depth, latitude, longitude) ;
        err_vo:_FillValue = 32767s ;
        err_vo:units = "m/s" ;
        err_vo:standard_name = "northward_sea_water_velocity_standard_error" ;
        err_vo:long_name = " Error on absolute surface geostrophic + depth Ekman velocity + tide: meridional component " ;
        err_vo:grid_mapping = "crs" ;
        err_vo:scale_factor = 0.001 ;
    short err_ue(time, depth, latitude, longitude) ;
        err_ue:_FillValue = 32767s ;
        err_ue:units = "m/s" ;
        err_ue:standard_name = "eastward_sea_water_velocity_due_to_ekman_drift_standard_error" ;
        string err_ue:long_name = "Error on depth Ekman driven velocity : zonal component" ;
        err_ue:grid_mapping = "crs" ;
        err_ue:scale_factor = 0.001 ;
    short err_ve(time, depth, latitude, longitude) ;
        err_ve:_FillValue = 32767s ;
```

```
        err_ve:units = "m/s" ;
        err_ve:standard_name = "northward_sea_water_velocity_due_to_ekman_drift_ standard_ error" ;
        string err_ve:long_name = "Error on depth Ekman driven velocity : meridional component" ;
        err_ve:grid_mapping = "crs" ;
        err_ve:scale_factor = 0.001 ;
    short err_ugos(time, depth, latitude, longitude) ;
        err_ugos:_FillValue = 32767s ;
        err_ugos:units = "m/s" ;
        err_ugos:standard_name = "surface_geostophic_eastward_sea_water_velocity_standard_error" ;
        err_ugos:long_name = "Error on absolute surface geostrophic velocity: zonal component " ;
        err_ugos:grid_mapping = "crs" ;
        err_ugos:scale_factor = 0.001 ;
    short err_vgos(time, depth, latitude, longitude) ;
        err_vgos:_FillValue = 32767s ;
        err_vgos:units = "m/s" ;
        err_vgos:standard_name = "surface_geostophic_northward_sea_water_velocity_standard_error" ;
        err_vgos:long_name = "Error on absolute surface geostrophic velocity: meridional component " ;
        err_vgos:grid_mapping = "crs" ;
        err_vgos:scale_factor = 0.001 ;
    short err_utide(time, depth, latitude, longitude) ;
        err_utide:_FillValue = 32767s ;
        err_utide:units = "m/s" ;
        err_utide:standard_name = "sea_water_x_velocity_due_to_tide_standard_error" ;
        err_utide:long_name = "Error on tide-induced velocity: zonal component " ;
        err_utide:grid_mapping = "crs" ;
        err_utide:scale_factor = 0.001 ;
    short err_vtide(time, depth, latitude, longitude) ;
        err_vtide:_FillValue = 32767s ;
        err_vtide:units = "m/s" ;
        err_vtide:standard_name = "sea_water_y_velocity_due_to_tide_velocity_standard_error" ;
        err_vtide:long_name = "Error on tide-induced velocity: meridional component " ;
        err_vtide:grid_mapping = "crs" ;
        err_vtide:scale_factor = 0.001 ;
// global attributes:
        :geospatial_lon_units = "degrees_east" ;
```

:geospatial_vertical_max = 0. ;

:product_version = "4.0" ;

:geospatial_lat_units = "degrees_north" ;

:creation_date = "2024-04-22 13:26:34" ;

:contact = "servicedesk.cmems@mercator-ocean.eu" ;

:references = "Mulet S., Rio M.-H., Etienne H., Artana C., Cancet M., Dibarboure G., Feng H., Husson R., Picot N., Provost C. and Strub P. T.: The new CNES-CLS18 global mean dynamic topography, Ocean Sci., 17, 789-808; Carrere L., Lyard F., Cancet M., Allain D., Fouchet E., Dabat. M, Tchilibou M., Ferrari R. and Faugere Y.: new FES2022 Tidal atlas, OSTST 2022." ;

:geospatial_vertical_min = -15. ;

:Conventions = "CF-1.6" ;

:institution = "CLS" ;

:lon_min = -179.875f ;

:lon_max = 179.875f ;

:lon_res = 0.25 ;

:lat_min = -89.875f ;

:lat_max = 89.875f ;

:lat_res = 0.25 ;

:title = "Monthly mean total surface and 15m velocities" ;